RE-ASSEMENT OF ELECTROACOUSTIC CHAIN

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ABSTRACT

The focus of my talk will be my work into the re-assessment of the traditional electroacoustic signal chain. This topic comprises the core of my Masters research at the NZSM.

As a basic model this chain can be defined as follows: sound source (within an acoustic space of some kind) to microphone, microphone signal to digital/analog workstation, then through speakers to the human ear via compression and rarefaction of air. It is my intention to reassess this signal chain by instrumentalising and physically manipulating elements in this chain that are generally considered to be static and invariant, such as microphone and loudspeakers as well as the acoustic environment in which sound is produced.
The context of my work emerges from a wider look at signal chains that have been manipulated, bent and broken to create new forms of music and sound. In a popular context, by including the hand *as instrument* into a record playback system (record, stylus, amp, speakers) turntablism and hip-hop were produced. In the context of sound-based arts there are examples ranging from Cage's prepared piano work, to Collins' hardware hacking and circuit bending. The creation of music from non- musical signal chains has also been accomplished (music created from amplified brain waves in Lucier's *'Music for Solo Performer'*).

Some of the techniques I will be employing within my work are; the movement of microphones around a sound source to create spatialisation and filtering within a stereo field and using the resonant bodies of instruments to enhance resonance of a piece of music (pace Lucier's work '*Music for piano with amplified sonorous vessels'* in which vessels are used as chambers to capture and enhance resonance of certain piano notes). Another technique I will be developing is the use of the speaker as an instrument. I will re-evaluate its use in electroacoustic music and exploit the physicalaspects of the speaker that will allow me to use it as a tool for transforming sound, rather than leaving it dormant in the signal chain (mentioned above). This may include covering the speaker to create filtering, moving it to create external spatialisation (Reich; '*Pendulum Music'*, Monahan; '*Speaker Swinging'*) that is not inherent in the aural material, or using it as a resonant body in which the vibrations of the structure itself are used as musical source material.

My talk will outline my reevaluation of models for production and reception of electroacoustic music and seek to develop new techniques to add to the signal chain or break it down to a point where the individual steps either merge or are totally isolated. In doing this, my aim is to move away from relying solely on digital signal processing in the studio to create music, but rather pushing the signal chain towards including tangible objects (such as the microphone, speaker and instruments) that will not only transform, but create and re-contextualise sound. Simply put, I aim to create methods of processing sound that exist on a physical level through approaching loudspeakers and microphones as instruments rather than just a means to capture and reproduce sound.